

```
reviews = [

"This product performs efficiently and meets expectations",

"The quality of the item is unsatisfactory",

"Delivery was completed on time and packaging was secure",

"The product stopped functioning after a short period",

"Good value for the price"]
```

```
selected = reviews[:3]

data = [d.lower() for d in selected]
```

```
!pip install transformers torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.12.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.10.22)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.20.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.5.1)
Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.29.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.12.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: click>=8.4.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0) (8.1.8)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0) (1.20.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0) (0.27.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0) (4.6.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0) (2025.8.3)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0) (3.10.1)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0) (0.14.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer->transformers) (2.18.0)
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->transformers) (0.1.2)
```

```
from transformers import BertTokenizer
import torch

tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:124: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>).
You will be able to reuse this secret in all of your notebooks.

```
for i, review in enumerate(data):

    encode = tokenizer(
        review,
        add_special_tokens=True,
        max_length=20,
        padding='max_length',
        truncation=True,
        return_tensors='pt'
    )
    input_ids = encode['input_ids'][0]

    token = tokenizer.convert_ids_to_tokens(input_ids)

    attention_mask = encode['attention_mask'][0]

    print(f"Tokens: \n{token}")
    print(f"Attention Mask: \n{attention_mask}")
    print(f"Input IDs: \n{input_ids}")
```

Tokens:
['[CLS]', 'this', 'product', 'performs', 'efficiently', 'and', 'meets', 'expectations', '[SEP]', '[PAD]', '[PAD]', '
Attention Mask:
tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0])
Input IDs:
tensor([101, 2023, 4031, 10438, 18228, 1998, 6010, 10908, 102, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0])
Tokens:
['[CLS]', 'the', 'quality', 'of', 'the', 'item', 'is', 'un', '##sat', '##is', '##factory', '[SEP]', '[PAD]', '[PAD]', '
Attention Mask:
tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0])
Input IDs:
tensor([101, 1996, 3737, 1997, 1996, 8875, 2003, 4895, 16846, 2483,
21450, 102, 0, 0, 0, 0, 0, 0, 0, 0])
Tokens:
['[CLS]', 'delivery', 'was', 'completed', 'on', 'time', 'and', 'packaging', 'was', 'secure', '[SEP]', '[PAD]', '[PAD]', '
Attention Mask:
tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0])
Input IDs:
tensor([101, 6959, 2001, 2949, 2006, 2051, 1998, 14793, 2001, 5851,
102, 0, 0, 0, 0, 0, 0, 0, 0, 0])

Start coding or [generate](#) with AI.

